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UNIVERSITY OF
BIRMINGHAM

Category Theory

Midlands Graduate School 2026

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Big Picture / Motivation

- **Category theory** is a “theory”, like **set theory**, like **group theory**, like **type theory**
- Types in functional programming $\leadsto$ sets (or set-like objects)
- Functions $\leadsto$ arrows/morphisms between types
- Category theory gives language to talk about **how** we compose computations, abstract patterns, and universal constructions that mirror programming constructs (pairs, function types, currying)
- It provides means of **abstraction** to reason and prove things **generically**, instead of on case-by-case basis

Course Outline

Part I: Sets and Warm-Up

Sets, functions, isomorphisms,
quotients and equivalences

Part II: Categories & Functors

Definitions and examples, products,
coproducts, commutative diagrams,
universal properties

Part III: Natural Transformations

Natural transformations, exponentials,
Yoneda Theorem

Part IV: Adjunctions & Free Objects

Adjunctions, free objects,
initial algebras, induction

These slides:

www.sergey-goncharov.org/mgs-2026



Recommended Sources

- **Steve Awodey** — *Category Theory*, 2nd ed., Oxford University Press, 2010
 - [accessible introduction]
- **Emily Riehl** — *Category Theory in Context*, Dover, 2016
 - [freely available at <https://math.jhu.edu/~eriehl/161/context.pdf>]
- **Roy Crole** — *Categories for Types*, Cambridge University Press, 1993
 - [CS-oriented; Category theory for types & denotational semantics]
- **Saunders Mac Lane** — *Categories for the Working Mathematician*, 2nd ed., Springer, 1998
 - [timeless classics]

Sets and Warm-Up



Why Begin with Sets?

- Mathematics builds on sets
- Types in programming follow set intuition
- Category theory (standardly) also builds on sets
- But: what exactly is a set?
 - OK, that is too ambitious – what is it roughly?
- We start with sets to discover something deeper is hiding underneath

Naïve (Philosophical) Sets

- Set is collection of objects, like

$$X = \left\{ \text{⚽}, \text{🐕}, \pi, \mathbb{N}, \text{♥} \right\}$$

- We write $x \in X$ to say that x is **member** of X , e.g. $\pi \in X$
- **Empty set** $\emptyset = \{ \}$ is a set of no elements
- There are various mechanisms to construct larger sets
- **Set comprehension**

$$\{x \in S \mid P(x)\} \quad \text{e.g.} \quad \text{evens} = \{n \in \mathbb{N} \mid \exists m \in \mathbb{N}. n = 2m\}$$

Intuition: Sets are containers

Finite Sets v.s. Finite Lists

- **(Finite) lists** over set X : $\text{List}(X) = \{[x_1, \dots, x_n] \mid n \in \text{Nat}, x_1 \in X, \dots, x_n \in X\}$
- **Notation:** $[]$ – **empty list**, $[x \mid xs]$ – list with **head** x and **tail** xs . E.g. $[x \mid []] = [x]$
- Finite sets are lists, where we additionally enable
 - **Commutativity:** $[\dots, x, y, \dots] \equiv [\dots, y, x, \dots]$
 - **Idempotence:** $[\dots, x, x, \dots] \equiv [\dots, x, \dots]$
- **(Finite) Multisets** = lists with commutativity, but not idempotence. Thus

Lists $\supseteq$ Multisets $\supseteq$ Sets

Sets (Slightly More) Formally

- Set theory postulates: everything is a set
- Only primitive relation between sets is ' $\in$ '

We then derive:

- **Natural numbers:** $0 = \emptyset$, $1 = \{0, \emptyset\} = \{\emptyset\}$, $2 = \{1, \emptyset\} = \{\{\emptyset\}, \emptyset\}, \dots$
- Set inclusion: $X \subseteq Y$ if $\forall x \in X. x \in Y$
 - **Extensionality axiom:** $X = Y$ if and only if $X \subseteq Y$ and $Y \subseteq X$
- **Powerset:** $Y \in \mathcal{P}(X)$ if and only if $Y \subseteq X$
- **Relations:** $R \subseteq X \times Y$
- **Functions:** $f: X \rightarrow Y$ – such relations $R \subseteq X \times Y$ that for every $x \in X$ there is unique $y \in Y$, such that $x R y$
 - So, in set theory: Sets $\leadsto$ Relations $\leadsto$ Functions

Empty Set and Singleton Sets

- No controversy about empty set: $\emptyset = \{ \}$ – set with no elements
- But what is one-element set? $\{\text{⚽}\}$? $\{\text{🐕}\}$? $\{\pi\}$?
- $\{\emptyset\}$ is good candidate
- For every set S , $\{S\}$ is also singleton set
- Singleton sets are characterized by property: they are sets with exactly one element
- $\{\emptyset\}$ is a concrete **representative** in the **class** of all singletons

Note on Classes

- All sets do not form a set, but a **class**

$\text{Set} < \text{Class}$

- Division into sets and proper classes is way to avoid paradoxes
- Other approach: **universes**

$\text{Set}_0 < \text{Set}_1 < \text{Set}_2 < \dots$

where the union of all sets in Set_i is in Set_{i+1}

- By saying “set” we can imagine set from Set_i , and by saying “class” we can imagine set from Set_{i+1}

Natural Numbers as Sets

- In computer science, we count from 0: $\mathbb{N} = \{0, 1, \dots\}$!
- **Von Neumann numerals:** $0 := \emptyset, n + 1 := n \cup \{n\}$ So:

$$0 = \emptyset, \quad 1 = \{\emptyset\}, \quad 2 = \{\emptyset, \{\emptyset\}\}, \quad 3 = \{\emptyset, \{\emptyset, \{\emptyset\}\}\}, \dots$$

This is very convenient, because $n + 1 = \{0, \dots, n\}$

- Alternatively **Zermelo numerals:** $0 := \emptyset, n + 1 := \{n\}$. This is less convenient

Ordered Pairs and Products

We would like to define **Cartesian products** $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$. But what is (x, y) ?

- **Kuratowski definition:** $(x, y) := \{\{x\}, \{x, y\}\}$
- Then (**PROVE IT**) :

$$(x, y) = (x', y') \iff x = x' \text{ and } y = y'.$$

- Using this, we can define:

$$X \times Y := \left\{ \{\{x\}, \{x, y\}\} \subseteq \mathcal{P}(X \cup Y) \mid x \in X, y \in Y \right\}$$

Pairs and Cartesian products are **constructed objects**!

Disjoint Unions

- **Union of sets:** $x \in X \cup Y$ iff $x \in X$ or $x \in Y$
- How to define **Disjoint Union** $X + Y (= X \uplus Y)$?
- Maybe $X + Y = \{(x, \bullet) \mid x \in X\} \cup \{(x, \blacktriangleright) \mid y \in Y\}$?

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- Maybe $X + Y = \{(x, \text{🍔}) \mid x \in X\} \cup \{(x, \text{🐘}) \mid y \in Y\}$?
- More seriously: $X + Y = \{(x, 0) \mid x \in X\} \cup \{(x, 1) \mid y \in Y\}$

Now compute with von Neumann numerals: $1 = \{0\}$, $2 = \{0, 1\}$. Then

$$1 + 1 = \{(1, 0), (1, 1)\} = \{\{\{1\}, \{1, 0\}\}, \{\{1\}\}\}$$

This is **not** equal to $2 = \{0, 1\}$! So

$$1 + 1 \neq 2$$

Functions between Sets

- Function $f: X \rightarrow Y$, assigns each $x \in X$ a unique $f(x) \in Y$
- Functions $f: X \rightarrow Y$ are special relations $f \subseteq X \times Y$
- Identity function $\text{id}_X: X \rightarrow X$, $\text{id}_X(x) = x$
- Equality of functions $f = g$ if for all x , $f(x) = g(x)$
 - This is provable principle (**PROVE IT**) , called **function extensionality**

Composition:

$$(g \circ f)(x) = g(f(x))$$

Proposition: $f \circ \text{id} = f$, $\text{id} \circ f = f$ and $f \circ (g \circ h) = (f \circ g) \circ h$ (associativity)

Bijection vs Isomorphism

Function $f: X \rightarrow Y$ is

- **Injection** if $f(x) = f(x')$ entails $x = x'$
- **Surjection** if for every $y \in Y$ there is $x \in X$, such that $f(x) = y$
- **Bijection** if it is both injection and surjection

Alternative view: $f: X \rightarrow Y$ is **isomorphism** if for some $f^{-1}: Y \rightarrow X$,

$$f \circ f^{-1} = \text{id}_Y \quad \text{and} \quad f^{-1} \circ f = \text{id}_X$$

Proposition: f is bijection if and only if f is isomorphism

Some Properties of Isomorphisms

It follows that (**PROVE IT**)

- f^{-1} is unique if exists
- If f is isomorphism, so is f^{-1}
- id is isomorphism and $\text{id}^{-1} = \text{id}$
- Composition of isomorphisms $f \circ g$ is isomorphism and $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

These are derivable from definition, without using that isomorphism = bijection!

Notation: $X \cong Y$ if there is isomorphism $f: X \rightarrow Y$

Up-to Isomorphism

- Big innovation of category theory: work with things via their descriptions
- We narrow down objects of interest as those that “fit best” to such description
- Such descriptions together with their best-fit condition are called **universal properties**
- They determine objects not uniquely, but **up-to isomorphism**

Example. A **singleton set** 1 is characterised by: there is exactly one function $X \rightarrow 1$ for every set X

- $\{\emptyset\}$, $\{\{\emptyset\}\}$, $\{\text{⚽}\}$ all satisfy this — and are all isomorphic
- Any two singletons are isomorphic, so we speak of **the singleton up to isomorphism**

Sets and Warm-Up



Equivalences and Quotients

Equivalence Relations

A relation $R \subseteq X \times X$ is an **equivalence relation** if

- **Reflexive**: $x R x$
- **Symmetric**: $x R y \Rightarrow y R x$
- **Transitive**: $x R y$ and $y R z \Rightarrow x R z$

For $x \in X$, define corresponding **equivalence class**

$$[x]_R = \{y \in X \mid x R y\}$$

Equivalence classes **partition** X : $X = \bigcup_{x \in X} [x]_R$, $[x]_R \cap [y]_R \neq \emptyset$ iff $x R y$

Quotient Sets

- Given equivalence relation $R \subseteq X \times X$, define **quotient set**

$$X/R = \{[x]_R \mid x \in X\}$$

- There is natural function

$$\pi: X \rightarrow X/R, \quad \pi(x) = [x]_R$$

- Two elements have same image iff they are related:

$$\pi(x) = \pi(y) \iff x R y$$

Example: **Integers** $\mathbb{Z} = (\mathbb{N} \times \mathbb{N}) / \sim$ where $(m, n) \sim (m', n')$ iff $m + n' = m' + n$ (**PROVE IT**)

Images and Inverse Images

Given $f: X \rightarrow Y$,

- For $A \subseteq X$ define **direct image**

$$f[A] = \{f(x) \mid x \in A\}$$

- For $B \subseteq Y$ define **inverse image**

$$f^{-1}[B] = \{x \in X \mid f(x) \in B\}$$

- Let $f^{-1}(x) = f^{-1}[\{x\}]$ – these are often called **fibers** (over $x \in Y$)

Equivalence Relations from Functions

- Given function $f: X \rightarrow Y$, **kernel equivalence** $\sim_f \subseteq X \times X$ is as follows:

$$x \sim_f y \iff f(x) = f(y)$$

This is indeed equivalence relation (**PROVE IT**)

- Equivalence classes are precisely the fibres

$$[x]_{\sim_f} = f^{-1}(f(x))$$

Surjections are Equivalent to Equivalences

- Start with equivalence $R \subseteq X \times X$, then with $\pi: X \rightarrow X/R$

$$R \cong \sim_\pi$$

— every equivalence is isomorphic to kernel equivalence of induced quotienting map

(**PROVE IT**)

- Start with surjection $f: X \rightarrow Y$, then

$$Y \cong X / \sim_f$$

— codomain of any surjection is isomorphic to quotient of domain by kernel equivalence

(**PROVE IT**)

Example. Let $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{Z}$, $f(m, n) = m - n$. Then:

$$(m, n) \sim_f (m', n') \iff m - n = m' - n' \iff m + n' = m' + n$$

— exactly the equivalence used to construct $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\sim$

Choice

When we say

*“Codomain of any surjection is isomorphic to
quotient of domain by kernel equivalence”*

there is a snag

To build isomorphism, we need **axiom of choice** to identify concrete x in equivalence class $[x]_{\sim_f}$

Without that, the equivalence becomes definition: it defines **exactly those** surjections that are quotient maps of equivalence relations

Category theory calls them **regular epimorphisms**

Categories & Functors



Natural Transformations



Adjunctions, Algebras & Coalgebras



<https://sergey-goncharov.org/mgs-2026>